MINGSHUAI CHEN

Formal Verification Group · College of Computer Science and Technology · Zhejiang University
Zheda Road 38, Hangzhou 310027, Zhejiang, China
☎ +86 18810578676 ✉ m.chen@zju.edu.cn 🌍 fiction-zju.github.io/author/mingshuai-chen

Employment

College of Computer Science and Technology, Zhejiang University Jan. 2023 - present
Assistant Professor of Computer Science leading the Formal Verification Group

- Deputy Director of the Office of Global Engagement, Zhejiang University (part-time, since Sep. 2026)
- Vice Dean of the Binjiang Institute of Zhejiang University (part-time, since Mar. 2026)

Dept. of Computer Science, RWTH Aachen University Sep. 2019 - Nov. 2022
Postdoctoral Researcher at Software Modeling and Verification Group *Head: Prof. Dr. Joost-Pieter Katoen*

Education

Institute of Software, Chinese Academy of Sciences Sep. 2013 - Jun. 2019
Ph.D. in Computer Science (with honour) at St. Key Lab. Comput. Sci. *Advisor: Prof. Dr. Naijun Zhan*

- Dissertation: Verification and Synthesis of Time-Delayed Dynamical Systems
- Award: CAS-President Special Award

Dept. of Computing Science, Carl von Ossietzky Universität Oldenburg Fall, 2015 - 2018
Visiting scholar at Hybrid Systems Group *Advisor: Prof. Dr. Martin Fränzle*

College of Computer Science and Technology, Jilin University Sep. 2009 - Jun. 2013
B.Sc. in Computer Science (with honour)

Research Interests

My primary research interest lies in the general scope of *formal verification* and *synthesis*, broadly construed in *mathematical logic* and *theoretical computer science*. I develop formal reasoning techniques for programs and hybrid discrete-continuous systems for ensuring the reliability and effectiveness of safety-critical cyber-physical systems, and aim to push the limits of automation as far as possible. My research interests include

- Logical Aspects of CS
- Formal Verification and Synthesis
- Programming Theory/Languages
- Cyber-Physical Systems
- Probabilistic/Quantum Systems
- AI for Formal Methods

Academic Services

Lecturing

The Principles of Compilers, B.Sc., ZJU SS 24/25/26

Teaching Assistant

Trends in Computer-Aided Verification, B.Sc./M.Sc., Seminar in Theoretical CS, RWTH Aachen SS 21/22

Concurrency Theory, M.Sc., RWTH Aachen WS 21-22

Probabilistic Programming, B.Sc./M.Sc., Seminar in Theoretical CS, RWTH Aachen WS 20-21

Theoretical Foundations of the UML, M.Sc., RWTH Aachen SS 20

Theories of Programming, M.Sc., UCAS WS 17-18/18-19

Editor & Committee Chair

Program Co-Chair of ICFEM 2027 Nov. 2027

Program Co-Chair of QEST+FORMATS 2026 Sep. 2026

Workshop Chair of ICFEM 2025 Nov. 2025

Guest Editor of Special Issue on Formal Methods and Their Applications of Journal of Software Nov. 2024

Co-Chair of Forum on Formal Methods and Their Applications at ChinaSoft 2024 — Nov. 2024

Co-Chair of Forum on Theor. Eng. Software under Uncertainty at ChinaSoft 2024 — Nov. 2024

Associate Editor of Frontiers in Computer Science (Theoretical Computer Science) — Mar. 2026 - present

Committee Member

Senior Member of the China Computer Federation (CCF) — Sep. 2024 - present

Executive Member of CCF Technical Committee on Formal Methods — Jul. 2023 - present

Member of CCF Technical Committee on Theoretical Computer Science — Jul. 2023 - present

Reviewer Panel of Mathematical Reviews — Oct. 2021 - present

Program/Review Committee Member of TACAS 2027/2026/2025, CAV 2026, OOPSLA 2026, FSTTCS 2026, LMPL@SPLASH 2026, SETSS 2026/2025, ICFEM 2025, ATVA 2025, TASE 2025, FSEN 2025, SETTA 2024, RTCSA 2024/2021, ICWS 2024 (SRG Workshop), SYNASC 2022 (Logic and Programming Track)

Repeatability/Artifact Evaluation Committee Member of ADHS 2021, VMCAI 2021

External Reviewer

POPL '27, FM '24, EMSOFT '24/19, VMCAI '24, TACAS '23/21, POPL '23, ICALP '22, TASE '22/15, SAFECOMP '22, CAV '21/20, ADHS '21/18, ECC '21/16, HSCC '20, FORMATS '20, RTSS '19, MEMOCODE '18, ATVA '18/15, ICECCS '17, TIME '16, VSTTE '16, UTP '16, ACM Trans. Softw. Eng. Methodol., J. Autom. Reason., ACM Trans. Cyber-Phys. Syst., Sci. Comput. Program., Form. Asp. Comput., Nonlinear Anal.: Hybrid Syst., J. Syst. Archit., C. Zhou's Festschrift, J.-P. Katoen's Festschrift, M. Fränzle's Festschrift, Royal Society Open Science

Honors & Awards

Recipient of the NSFC Excellent Young Scientists Fund Program (Overseas) — Oct. 2023

High-Impact Publication [43] in CS by Chinese Researchers across from Springer Nature — Feb. 2021

Nomination for the CAS Excellent Doctoral Dissertation Award [1] — Mar. 2020

CAS-President Special Award (1st awardee from ISCAS ever since its inception in 1985) — Jul. 2019

Best Paper Award [43] at FMAC 2019 — Dec. 2019

Distinguished Paper Award [46] at ATVA 2018 — Oct. 2018

National Scholarship — Oct. 2018/2010

Selected Attendee of the 6th Heidelberg Laureate Forum — Sep. 2018

Outstanding Student Award of UCAS Scientific Research Project — Dec. 2013

Grants

[PI] NSFC General Program: Fixed-Point Theories for Quantitative Verifi. & Symb. Inference — 2026 - 2029

[PI] NSFC Excellent Young Scientists Fund (Overseas): Formal Design of Safety-Critical CPS — 2024 - 2027

[PI] ZJNSF Major Program: Hybrid Comput. Models and Theories for Heterogeneous Comput. — 2024 - 2027

[PI] CASC Open Fund: Test Generation and Prioritization for Aerospace Embedded Software — 2025 - 2026

[PI] CCF-Huawei Populus Grove Fund: Modular Verification via LLMs + Symbolic Reasoning — 2025 - 2026

[PI] Huawei Tech-Cooperation: LLM-Driven Verification for Large-Scale Programs — 2025 - 2026

[Co-I] ZJNSF Major Program: Theory of Generative Operating Systems — 2024 - 2027

[Co-I] ZJDST R&D Program: Adaptive Operating Systems Against Complex Environments — 2024 - 2027

[Co-I] ZJDST R&D Program: Quantum Key Distribution Systems: Terminals and Protocols — 2024 - 2027

[PI] CCF-Huawei Populus Grove Fund: Self-Adaptive Abstract Interpretation for C Programs — 2023 - 2024

[PI] ZJU 100 Young Professor: Foundations of Cyber-Physical Systems		2023 - 2029
[PI] Qizhen Scholar: Talent program funded by ZJU Education Foundation		2023 - 2026
[Co-I] National Key R&D Program: General Theory and Tech. of Service Intelligent Supervision		2022 - 2025
[Co-I] NSFC General Program: Formal Verification of Delayed Dynamical and Hybrid Systems		2019 - 2022

Selected Publications

cf. page 6 for a complete list of publications

Inf. Com. On termination of polynomial programs with equality conditions
Y. Li, M. Chen✉, L. Zhao, N. Zhan, H. Lu, G. Wu and J.-P. Katoen
Information and Computation, 2026

OOPSLA '26 A Tale of 1001 LoC: Potential Runtime Error-Guided Specification Synthesis
Z. Wang, T. Lin, M. Chen✉, H. Li, M. Yang, X. Yi, S. Qin, Y. Luo, X. Li, B. Gu, L. Lu, and J. Yin
The OOPSLA 2026 issue of the Proc. of the ACM on Programming Languages (PACMPL)

OOPSLA '26 Formalizing the Linux eBPF Core ISA: A Mechanized Operational Semantics
S. Yuan, Y. Tang, T. Cao, F. Besson, J.-P. Talpin, and M. Chen✉
The OOPSLA 2026 issue of the Proc. of the ACM on Programming Languages (PACMPL)

HSCC '26 Derivative-agnostic inference of nonlinear hybrid systems
H. Yu, B. Ma, M. Chen✉, H. Dong, J. An, B. Gu, N. Zhan, and J. Yin.
29th ACM International Conference on Hybrid Systems: Computation and Control (HSCC 2026)

CAV '25 On the Almost-Sure Termination of Probabilistic Counter Programs
S. Novozhilov, M. Yang, M. Chen✉, Z. Li, and J. Yin
37th Int. Conf. on Computer Aided Verification (CAV 2025)

OOPSLA '24 Exact Bayesian Inference for Loopy Probabilistic Programs
L. Klinkenberg, C. Blumenthal, M. Chen✉, D. Haase, and J.-P. Katoen
The OOPSLA 2024 issue of the Proc. of the ACM on Programming Languages (PACMPL)

ASE '24 Parf: Adaptive Parameter Refining for Abst. Interp. **2nd Prize at ChinaSoft 2024**
Z. Wang, L. Yang, M. Chen✉, Y. Bu, Z. Li, Q. Wang, S. Qin, X. Yi, and J. Yin
39th IEEE/ACM Int. Conf. on Automated Software Engineering (ASE 2024)

FM '24 Proving Functional Program Equivalence via Directed Lemma Synthesis
Y. Sun, R. Ji, J. Fang, X. Jiang, M. Chen, and Y. Xiong
26th Int. Symp. on Formal Methods (FM 2024)

OOPSLA '23 Lower Bounds for Possibly Divergent Probabilistic Programs
S. Feng, M. Chen✉, H. Su, B. L. Kaminski, J.-P. Katoen, and N. Zhan
The OOPSLA 2023 issue of the Proc. of the ACM on Programming Languages (PACMPL)

TACAS '23 Probabilistic Program Verification via Inductive Synthesis of Inductive Invariants
K. Batz, M. Chen✉, S. Junges, B. L. Kaminski, J.-P. Katoen, and C. Matheja
29th Int. Conf. on Tools and Algorithms for Construction and Analysis of Systems (TACAS 2023)

Inf. Com. Encoding Inductive Invariants as Barrier Certificates
Q. Wang, M. Chen✉, B. Xue, N. Zhan, and J.-P. Katoen
Information and Computation, 2022

CAV '22 Does a Program Yield the Right Distribution?
M. Chen✉, J.-P. Katoen, L. Klinkenberg, and T. Winkler
34th Int. Conf. on Computer Aided Verification (CAV 2022)

Acta Inf. Indecision and Delays Are the Parents of Failure
M. Chen✉, M. Fränzle, Y. Li, P. N. Mosaad, and N. Zhan
Acta Informatica, 2021

CAV '21 Latticed k-Induction with an Application to Probabilistic Programs
K. Batz, M. Chen✉, B. L. Kaminski, J.-P. Katoen, C. Matheja, and P. Schröer
33rd Int. Conf. on Computer Aided Verification (CAV 2021)

CAV '21 Synthesizing Invariant Barrier Certificates via Difference-of-Convex Programming
Q. Wang, M. Chen✉, B. Xue, N. Zhan, and J.-P. Katoen
33rd Int. Conf. on Computer Aided Verification (CAV 2021)

CAV '20 Unbounded-Time Safety Verification of Stochastic Differential Dynamics
S. Feng, M. Chen✉, B. Xue, S. Sankaranarayanan, and N. Zhan
32nd Int. Conf. on Computer Aided Verification (CAV 2020)

TACAS '20 Learning One-Clock Timed Automata — **Best Paper Award at FMAC 2019**
J. An, M. Chen, B. Zhan, N. Zhan, and M. Zhang
26th Int. Conf. on Tools and Algorithms for Construction and Analysis of Systems (TACAS 2020)

CAV '19 Taming Delays in Dynamical Systems
S. Feng, M. Chen✉, N. Zhan, M. Fränzle, and B. Xue
31st Int. Conf. on Computer Aided Verification (CAV 2019)

CADE '19 NIL: Learning Nonlinear Interpolants
M. Chen✉, J. Wang, J. An, B. Zhan, D. Kapur, and N. Zhan
27th Int. Conf. on Automated Deduction (CADE 2019)

ATVA '18 What's to Come Is Still Unsure — **Distinguished Paper Award**
M. Chen✉, M. Fränzle, Y. Li, P. N. Mosaad, and N. Zhan
16th Int. Symp. on Automated Technology for Verification and Analysis (ATVA 2018)

IEEE TAC Reachability Analysis for Solvable Dynamical Systems
T. Gan, M. Chen, Y. Li, B. Xia, and N. Zhan
IEEE Trans. Automat. Contr., 2018

IJCAR '16 Interpolant Synthesis for Quadratic Polynomial Inequalities and Combination with *EUF*
T. Gan, L. Dai, N. Zhan, D. Kapur, and M. Chen
8th Int. Joint Conf. on Automated Reasoning (IJCAR 2016)

FM '16 Validated Simulation-Based Verification of Delayed Differential Dynamics
M. Chen, M. Fränzle, Y. Li, P. N. Mosaad, and N. Zhan
21st Int. Symp. on Formal Methods (FM 2016)

Selected Tools/Prototypes

• **Preguss**: A tool for inferring formal specifications that integrates static analysis and deductive verification in a sound and terminating manner. Preguss is, to the best of our knowledge, the first automated tool capable of proving RTE-freeness of real-world programs with over 1,000 LoC (with marginal human effort).

• **Parf**: A tool geared towards automatically refining abstraction strategies of static analyzers for C programs; it has been awarded the 2nd prize (ranking 5/74) in the Software Prototype Competition as part of ChinaSoft 2024 and applied in Huawei's key product lines such as wireless base stations and cryptographic libraries.

• **Prodigy**: A tool that decides whether a given probabilistic loop agrees with an (invariant) specification encoded as a loop-free program; it supports exact inference and efficient queries on posterior distributions.

• **cegispro2**: A tool that proves upper- and/or lower bounds on expected outcomes of possibly infinite-state probabilistic programs by synthesizing piecewise linear quantitative inductive invariants.

• **kipro2**: A tool that performs in parallel latticed k-induction and BMC to fully automatically verify upper bounds on expected values of possibly infinite-state probabilistic programs.

• **BMI-DC**: A prototype that proves unbounded-time safety of differential dynamical systems by synthesizing invariant barrier certificates via difference-of-convex programming.

• **NIL**: A learning-based tool that automatically synthesizes non-trivial (reverse) Craig interpolants for the quantifier-free theory of nonlinear arithmetic.

• **DGame**: A tool that automatically synthesizes finite-memory controllers (a.k.a. winning strategies) for safety games under delayed information.

• **MPPs**: A procedure for deciding termination by computing the set of non-terminating inputs for multi-path polynomial programs with equality conditions.

• **MARS**: A toolchain for modelling, analyzing and verifying hybrid systems, which has been successfully applied in the verification of the Chinese lunar lander Chang'e-3 and the high-speed rail in China.

Selected Talks

Dissertation Defence

• Verification & Synthesis of Time-Delayed Dynamics, *Doctoral Dissertation Defence at ISCAS*, Beijing, China, May 2019.

Tutorials

• Taming Delays in Cyber-Physical Systems, *ESWEEK*, Shanghai, China, Oct. 2022. [Tutorial co-presented with Naijun Zhan].

• Formal Analysis, Verification and Design of Safety-Critical CPS, *RTSS*, Houston, USA, Dec. 2020. [Tutorial co-presented with Lei Bu, Qixin Wang and Naijun Zhan].

Conferences, Workshops, Seminars & Visits

• Derivative-Agnostic Inference of Nonlinear Hybrid Systems, *International Online Seminar on Interval Methods in Control Engineering*, Apr. 2025.

• Parf: An Adaptive Abstraction-Strategy Refiner for Static Analysis, *Dishui Lake Forum*, *East China Normal University*, Shanghai, China, Jan. 2025.

• Reasoning about Loopy Probabilistic Programs, *PurPL Seminar*, *Purdue University*, West Lafayette, USA, Sep. 2024 & *Colloquium Talk at Seminar on Software Engineering and Programming Languages*, *Hong Kong University of Science and Technology*, Hong Kong, China, Jan. 2024 & *CCF Beautiful Lake Seminars on Formal Verification of Large-Scale Infrastructure Software*, Suzhou, China, Aug. 2024.

• Lower Bounds for Possibly Divergent Probabilistic Programs, *MOVES' Colloquium at Schloss Dagstuhl*, Saarbrücken, Germany, Apr. 2022 & *ROCKS*, Nijmegen, Netherlands, May 2022 & *OOPSLA*, Lisbon, Portugal, Oct. 2023.

• Latticed k-Induction with an Application to Probabilistic Programs, *Information Sciences Seminar* & *Youth Forum*, *Peking University*, Beijing, China, Oct. 2021 & *ISCAS Seminar*, *ISCAS*, Beijing, China, Dec. 2022.

• Synthesizing Invariant Barrier Certificates via Difference-of-Convex Programming, *Seminar on Cyber-Physical Systems*, *University of Southampton*, Southampton, UK, Apr. 2021 & *CAV*, Los Angeles, USA, Jul. 2021.

• On ∞-Safety of Stochastic Differential Dynamics, *MOVES Seminar*, *RWTH Aachen University*, Aachen, Germany, Apr. 2020 & *CAV*, Los Angeles, USA, Jul. 2020.

• (In-)Variant Synthesis for Probabilistic Programs [Immature Idea], *MOVES' Winter Colloquium at Kleinwalsertal*, Kleinwalsertal, Austria, Feb. 2020.

• Interpolation over Nonlinear Arithmetic - Towards Program Reasoning and Verification, *PKU Seminar on Programming Languages*, *Peking University*, Beijing, China, Sep. 2019 & *MOVES Seminar*, *RWTH Aachen University*, Aachen, Germany, Nov. 2019 & *FACAS*, La Falda, Córdoba, Argentina, Mar. 2022.

• Taming Delays in Dynamical Systems - Unbounded Verification of Delay Differential Equations, *CAV*, New York City, USA, Jul. 2019.

• Modelling · Verification · Synthesis - A Peek into the Blueprint of Hybrid Systems, *RWTH Aachen University*, Aachen & *Technische Universität München*, München, Germany, Oct. 2018.

• What's to Come is Still Unsure - Synthesizing Controllers Resilient to Delayed Interaction, *ATVA*, Los Angeles, USA, Oct. 2018 & *CAP*, Beijing, China, Sep. 2018 & *MISSION@INVAP*, San Carlos de Bariloche, Argentina, Feb. 2022.

• Towards Delays in Dynamical and Control Systems - Verification & Synthesis, *Universität des Saarlandes*, Saarbrücken, Germany, Jul. 2016 & *LEDS*, Shanghai, China, Dec. 2016.

• Validated Simulation-Based Verification of Delayed Differential Dynamics, *FM*, Limassol, Cyprus, Nov. 2016.

• Computing Reachable Sets of Linear Vector Fields Revisited, *ECC*, Aalborg, Denmark, Jun. 2016.

• A Two-Way Path between Formal and Informal Design of Embedded Systems, *UTP* & *iFM*, Reykjavík, Iceland, Jun. 2016.

• HHL Prover: An Improved Interactive Theorem Prover for Hybrid Systems, *ICFEM*, Paris, France, Nov. 2015.

• Decidability of Reachability for a Family of Linear Vector Fields, *ATVA*, Shanghai, China, Oct. 2015.

Visits & Participations

Academic Visits

ALPACAS Research Group, Hong Kong University of Science and Technology, Hong Kong, China Jan. 2024

Software Modeling and Verification Group, RWTH Aachen University, Aachen, Germany Oct. 2023/2018

St. Key Lab. Comput. Sci., Institute of Software, Chinese Academy of Sciences, Beijing, China Dec. 2022

Hybrid Systems Group, C. v. Ossietzky Universität Oldenburg, Oldenburg, Germany Fall, 2015 - 2018

Chair of Robotics, AI and Real-time Syst., Technische Universität München, München, Germany Oct. 2018

Dependable Systems and Software Group, Universität des Saarlandes, Saarbrücken, Germany Jul. 2016

Conferences & Workshops

CAV '25, ChinaSoft '24, SPLASH '23, ESWEEK '22, ROCKS '22, FACAS '22, MISSION '22, RP '21, LPP '21, CPS-IoT Week '21, VSOW03 '21, CAV '21/20/19/17, RTSS '20, ETAPS '20, ATVA '18/15, HLF '18, CONFESTA '18, CAP '18/17, SETTA '17, FMAC '17/16, FM '16, ECC '16, iFM&UTP '16, LEDS '16/14, ICFEM '15, CDZ '14, SAVE '14, Dagstuhl Seminars (No. 22154), Shonan Meeting (No. 237), Beautiful Lake Seminars (No. 18)

Summer/Autumn Schools

The Summer School on Formal Methods, Beijing, China Aug. 2019/2018

The 3rd School on Engineering Trustworthy Software Systems, Chongqing, China Apr. 2017

The 2nd AVACS Autumn School, Oldenburg, Germany Oct. 2015

The 4th Summer School in Symbolic Computation, Beijing, China Aug. 2015

The 5th Summer School on Formal Techniques, Atherton, CA, USA May 2015

The 4th SAT/SMT Summer School, Semmering, Austria Jul. 2014

Languages

English: working-language German: for survival only Chinese: mother-tongue

Complete List of Publications

Dissertation

[1] M. Chen✉. Verification and synthesis of time-delayed dynamical systems. *PhD Dissertation*, Institute of Software, Chinese Academy of Sciences, China, 2019 (in Chinese). [Nomination for the CAS Excellent Doctoral Dissertation Award].

Book Chapters

[2] Z. Li, M. Yang, S. Feng, and M. Chen✉. Fixed-point reasoning for stochastic systems: A survey of recent advancements and open challenges. In *Design and Verification of Cyber-Physical Systems: From Theory to Applications*, pages 101–125. Springer, 2026.

[3] S. Feng, T. Yang, M. Chen✉, and N. Zhan. A unified framework for quantitative analysis of probabilistic programs. In *Principles of Verification: Cycling the Probabilistic Landscape*, pages 230–254. Springer, 2025.

[4] M. Chen✉, X. Han, T. Tang, S. Wang, M. Yang, N. Zhan, H. Zhao, and L. Zou. MARS: A toolchain for modelling, analysis and verification of hybrid systems. In *Provably Correct Systems*, pages 39–58. Springer, 2017.

Journal Articles

[5] Z. Zhang, R. Chang, M. Chen, W. Shen, C. Yu, H. Huang, Q. Dai, and Y. Zhao. PA-Boot: A formally verified authentication protocol for multiprocessor secure boot under hardware supply-chain attacks. *IEEE Trans. Inf. Forensics Secur.*, 21:476–489, 2026.

[6] N. Zhan, J. Woodcock, J. Wang, and M. Chen✉. A brief history of formal methods in China. *Form. Asp. Comput.*, 38(3), 2026.

[7] S. Yuan, Y. Tang, T. Cao, F. Besson, J.-P. Talpin, and M. Chen✉. Formalizing the Linux eBPF core ISA: A mechanized operational semantics and its real-world applications. *Proc. ACM Program. Lang.*, number OOPSLA2, (OOPSLA2), 2026. [To appear].

[8] Z. Wang, T. Lin, M. Chen✉, H. Li, M. Yang, X. Yi, S. Qin, Y. Luo, X. Li, B. Gu, L. Lu, and J. Yin. A tale of 1001 LoC: Potential runtime error-guided specification synthesis for verifying large-scale programs. *Proc. ACM Program. Lang.*, 10(OOPSLA1), 2026. [Arfifact Evaluated].

[9] Y. Li, M. Chen✉, L. Zhao, N. Zhan, H. Lu, G. Wu, and J.-P. Katoen. On termination of polynomial programs with equality conditions. *Inf. Comput.*, 312:105487, 2026.

[10] S. Feng, H. Su, H. Wu, J. An, M. Chen✉, and N. Zhan. Tolerant barrier certificates for stochastic systems. *IEEE Trans. Comput. Aided Des. Integr. Circuits Syst.*, 2026.

[11] D. Xiang, L. Lu, S. Tan, X. Jia, Z. Zhou, G. Sun, M. Chen, and J. Yin. AdaptDQC: Adaptive distributed quantum computing with quantitative performance analysis. *IEEE Trans. Comput.*, 74(10):3277–3290, 2025.

[12] Z. Wang, M. Chen✉, T. Lin, L. Yang, J. Zhuo, Q. Wang, S. Qin, X. Yi, and J. Yin. Parf: An adaptive abstraction-strategy tuner for static analysis. *J. Comput. Sci. Technol.*, 40(4):993–1005, 2025.

[13] S. Tan, L. Lu, C. Lang, M. Chen, and J. Yin. Fast-USYN: Fast synthesis from unitary matrices to high-quality quantum circuits. *J. Softw.*, 36(8):3431, 2025. (in Chinese).

[14] J. Chen, F. Wang, S. Pang, M. Chen✉, M. Xi, T. Zhao, and J. Yin. A privacy policy text compliance reasoning framework with large language models for healthcare services. *Tsinghua Sci. Tech.*, 30(4):1831–1845, 2025.

[15] L. Klinkenberg, C. Blumenthal, M. Chen✉, D. Haase, and J. Katoen. Exact Bayesian inference for loopy probabilistic programs. *Proc. ACM Program. Lang.*, 8(OOPSLA1):923–953, 2024. [Arfifact Evaluated].

[16] S. Feng, M. Chen✉, H. Su, B. L. Kaminski, J.-P. Katoen, and N. Zhan. Lower bounds for possibly divergent probabilistic programs. *Proc. ACM Program. Lang.*, 7(OOPSLA1):696–726, 2023.

[17] Q. Wang, M. Chen✉, B. Xue, N. Zhan, and J.-P. Katoen. Encoding inductive invariants as barrier certificates: Synthesis via difference-of-convex programming. *Inf. Comput.*, 289:104965, 2022.

[18] M. Chen✉, M. Fränzle, Y. Li, P. N. Mosaad, and N. Zhan. Indecision and delays are the parents of failure - Taming them algorithmically by synthesizing delay-resilient control. *Acta Informatica*, 58(5):497–528, 2021.

[19] J. Wang, J. An, M. Chen, N. Zhan, L. Wang, M. Zhang, and T. Gan. From model to implementation: A network-algorithm programming language. *Sci. China Inf. Sci.*, 63(7), 2020.

[20] M. Fränzle, M. Chen, and P. Kröger. In memory of Oded Maler: Automatic reachability analysis of hybrid-state automata. *ACM SIGLOG News*, 6(1):19–39, 2019.

[21] T. Gan, M. Chen, Y. Li, B. Xia, and N. Zhan. Reachability analysis for solvable dynamical systems. *IEEE Trans. Automat. Contr.*, 63(7):2003–2018, 2018.

Peer-Reviewed Conference Papers

[22] H. Yu, B. Ma, M. Chen✉, H. Dong, J. An, B. Gu, N. Zhan, and J. Yin. Derivative-agnostic inference of nonlinear hybrid systems. In *Proc. of HSCC '26*, 2026. [To appear].

[23] Y. Tang, S. Yuan, J.-P. Talpin, and M. Chen✉. Formal specification of Linux eBPF instruction set architecture in Sail. In *Proc. of SOSP (eBPF) '26*, 2026. [To appear].

[24] K. Zhou, L. Lu, D. Xiang, C. Tao, A. Wu, J. Leng, F. Liu, M. Chen, and J. Yin. Vegapunk: Accurate and fast decoding for quantum ldpc codes with online hierarchical algorithm and sparse accelerator. In *Proc. of MICRO '25*, page 719–732. ACM, 2025.

[25] Z. Wang, T. Lin, M. Chen✉, M. Yang, H. Li, X. Yi, S. Qin, and J. Yin. Preguss: It analyzes, it specifies, it verifies. In *Proc. of LMPL '25*, page 118–123. ACM, 2025.

[26] W. Tian, L. Lu, S. Tan, S. Li, H. Li, T. Chu, X. Zhang, M. Chen, and J. Yin. YOUTIAO: Hybrid multiplexing with dynamic qubit grouping for low-cost and scalable quantum wiring. In *Proc. of MICRO '25*, page 595–608. ACM, 2025.

[27] Y. Sun, M. Chen✉, T. Zhao, K. Zhao, H. Li, J. Chen, Z. Wang, L. Lu, X. Zhao, S. Deng, and J. Yin. Horae: A domain-agnostic language for automated service regulation. In *Proc. of IJCAI '25*, pages 9349–9356. International Joint Conferences on Artificial Intelligence Organization, 2025.

[28] Y. Sun, M. Chen✉, T. Zhao, R. Xu, Z. Zhang, and J. Yin. The self-improvement paradox: Can language models bootstrap reasoning capabilities without external scaffolding? In *Proc. of ACL '25 (Findings)*, pages 6501–6512, 2025.

[29] S. Novozhilov, M. Yang, M. Chen✉, Z. Li, and J. Yin. On the almost-sure termination of probabilistic counter programs. In *Proc. of CAV '25*, pages 82–104, 2025. [Arfifact Evaluated].

[30] M. Yang, K. Batz, M. Chen, J.-P. Katoen, Z. Wu, and J. Yin. Latticed craig interpolation with an application to probabilistic verification. In *CIBD '24*, 2024. [Extended Abstract].

[31] Z. Wang, L. Yang, M. Chen✉, Y. Bu, Z. Li, Q. Wang, S. Qin, X. Yi, and J. Yin. Parf: Adaptive parameter refining for abstract interpretation. In *Proc. of ASE '24*, pages 1082–1093. ACM, 2024.

[32] S. Tan, H. Zhang, J. Yu, C. Lang, Y. Shang, X. Zhao, M. Chen, Y. Liang, L. Lu, and J. Yin. QuFEM: Fast and accurate quantum readout calibration using the finite element method. In *Proc. of ASPLOS '24*, pages 948–963. ACM, 2024. [To appear].

[33] S. Tan, D. Xiang, L. Lu, J. Lu, Q. Jiang, M. Chen, and J. Yin. MorphQPV: Exploiting isomorphism in quantum programs to facilitate confident verification. In *Proc. of ASPLOS '24*, pages 671–688. ACM, 2024.

[34] Y. Sun, R. Ji, J. Fang, X. Jiang, M. Chen, and Y. Xiong. Proving functional program equivalence via directed lemma synthesis. In *Proc. of FM '24*, page 538–557, 2024. [Arfifact Evaluated].

[35] Y. Sun, M. Chen⊠, K. Zhao, and J. Chen. Horae: A domain-agnostic modeling language for automating multimodal service regulation. In *Proc. of ICWS (SRG) '24*, 2024. [Work-in-Progress Paper].

[36] J. Chen, F. Wang, S. Pang, S. Tan, M. Chen⊠, T. Zhao, M. Xi, and J. Yin. UniGM: Unifying multiple pre-trained graph models via adaptive knowledge aggregation. In *Proc. of MM '24*, pages 8556–8565. ACM, 2024.

[37] L. Klinkenberg, T. Winkler, M. Chen, and J.-P. Katoen. Exact probabilistic inference using generating functions. In *LAFI '23*, 2023. [Extended Abstract].

[38] K. Batz, M. Chen⊠, S. Junges, B. L. Kaminski, J.-P. Katoen, and C. Matheja. Probabilistic program verification via inductive synthesis of inductive invariants. In *Proc. of TACAS '23*, pages 410–429, 2023. [Arfifact Evaluated].

[39] M. Chen⊠, J.-P. Katoen, L. Klinkenberg, and T. Winkler. Does a program yield the right distribution? Verifying probabilistic programs via generating functions. In *Proc. of CAV '22*, pages 79–101, 2022. [Arfifact Evaluated].

[40] Q. Wang, M. Chen⊠, B. Xue, N. Zhan, and J.-P. Katoen. Synthesizing invariant barrier certificates via difference-of-convex programming. In *Proc. of CAV '21*, pages 443–466, 2021. [Arfifact Evaluated].

[41] K. Batz, M. Chen⊠, B. L. Kaminski, J.-P. Katoen, C. Matheja, and P. Schröer. Latticed k-induction with an application to probabilistic programs. In *Proc. of CAV '21*, pages 524–549, 2021. [Arfifact Evaluated].

[42] S. Feng, M. Chen⊠, B. Xue, S. Sankaranarayanan, and N. Zhan. Unbounded-time safety verification of stochastic differential dynamics. In *Proc. of CAV '20*, pages 327–348, 2020. [Arfifact Evaluated].

[43] J. An, M. Chen, B. Zhan, N. Zhan, and M. Zhang. Learning one-clock timed automata. In *Proc. of TACAS '20*, pages 444–462, 2020. [Arfifact Evaluated · Best Paper Award at FMAC 2019].

[44] S. Feng, M. Chen⊠, N. Zhan, M. Fränzle, and B. Xue. Taming delays in dynamical systems - Unbounded verification of delay differential equations. In *Proc. of CAV '19*, pages 650–669, 2019. [Arfifact Evaluated].

[45] M. Chen⊠, J. Wang, J. An, B. Zhan, D. Kapur, and N. Zhan. NIL: Learning nonlinear interpolants. In *Proc. of CADE '19*, pages 178–196, 2019.

[46] M. Chen⊠, M. Fränzle, Y. Li, P. N. Mosaad, and N. Zhan. What's to come is still unsure - Synthesizing controllers resilient to delayed interaction. In *Proc. of ATVA '18*, pages 56–74, 2018. [Distinguished Paper Award].

[47] B. Xue, P. N. Mosaad, M. Fränzle, M. Chen, Y. Li, and N. Zhan. Safe over- and under-approximation of reachable sets for delay differential equations. In *Proc. of FORMATS '17*, pages 281–299, 2017.

[48] T. Gan, L. Dai, B. Xia, N. Zhan, D. Kapur, and M. Chen. Interpolant synthesis for quadratic polynomial inequalities and combination with *EUF*. In *Proc. of IJCAR '16*, pages 195–212, 2016.

[49] T. Gan, M. Chen, Y. Li, B. Xia, and N. Zhan. Computing reachable sets of linear vector fields revisited. In *Proc. of ECC '16*, pages 419–426, 2016.

[50] M. Chen, A. P. Ravn, S. Wang, M. Yang, and N. Zhan. A two-way path between formal and informal design of embedded systems. In *Proc. of UTP '16*, pages 65–92, 2016.

[51] M. Chen, M. Fränzle, Y. Li, P. N. Mosaad, and N. Zhan. Validated simulation-based verification of delayed differential dynamics. In *Proc. of FM '16*, pages 137–154, 2016.

[52] T. Gan, M. Chen, L. Dai, B. Xia, and N. Zhan. Decidability of the reachability for a family of linear vector fields. In *Proc. of ATVA '15*, pages 482–499, 2015.